# EnvisionObjectAnnotator - MobileSAM+Cutie evaluation

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## 1 Summary

This report presents a quantitative analysis of the performance of EnvisionObjectAnnotator using MobileSAM+Cutie for its backend. MobileSAM uses a lightweight image encoder making it ideal for efficiency; in the workflow, the user creates an object and adds a point, and MobileSAM generates an accurate mask around the object in almost real-time. These masks are passed to Cutie, which propagates and tracks them throughout the scene. This lightweight nature makes it much faster than SAM2, while still maintaining a competitive performance.

The MobileSAM+Cutie implementation can be found in this fork of the original repository, there are some videos embedded in the README that show an example. Note that this is not a very formalized or polished version, but rather a result of some personal backend comparison testing.

## 2 SAM2 vs Cutie comparison

The table below shows the results of using this new backend on the exact same data. Notably, there is a large gain in recall, at the cost of precision and slightly of specificity. This likely means that Cutie is more sensitive to over-detection, whereas SAM2 seemed to have a problem of under-detection.

| | SAM2 | | | MobileSAM+Cutie | | |
|---|---|---|---|---|---|---|
| Group | Precision | Recall | Specificity | Precision | Recall | Specificity |
| P06 Caregiver | 63.0 | 41.6 | 97.3 | 45.3 | 39.3 | 94.7 |
| P06 Child | 71.9 | 13.5 | 99.3 | 55.0 | 36.3 | 95.9 |
| P101 Caregiver | 32.5 | 2.7 | 99.1 | 53.7 | 27.0 | 96.4 |
| P101 Child | 78.8 | 9.2 | 99.8 | 67.1 | 51.9 | 97.7 |
| **Overall** | **63.4** | **18.0** | **98.8** | **53.8** | **37.6** | **96.1** |

**Table 1:** Precision, recall, and specificity of the automated gaze annotation, comparing the SAM2 pipeline against the MobileSAM+Cutie backend.

Given these metrics, we can calculate the F1 score, which is the harmonic mean of precision and recall, generally used to compare the performance of classification models, with $F_1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$, and thus:

$F_1\text{SAM} = \frac{2 \times 0.634 \times 0.180}{0.634 + 0.180} = 0.280$

$F_1\text{Cutie} = \frac{2 \times 0.538 \times 0.376}{0.538 + 0.376} = 0.443$

Cutie achieves a higher F1 score; this indicates that even though precision dropped slightly, the gain of recall was substantially more beneficial.

The next section presents the same quantitative analysis that was done in the previous report, including confusion matrices and object-level comparisons per dyad.

# 3 Quantitative analysis

## 3.1 Confusion matrices

### *3.1.1 Dyad 1: P06S2*

Caregiver:

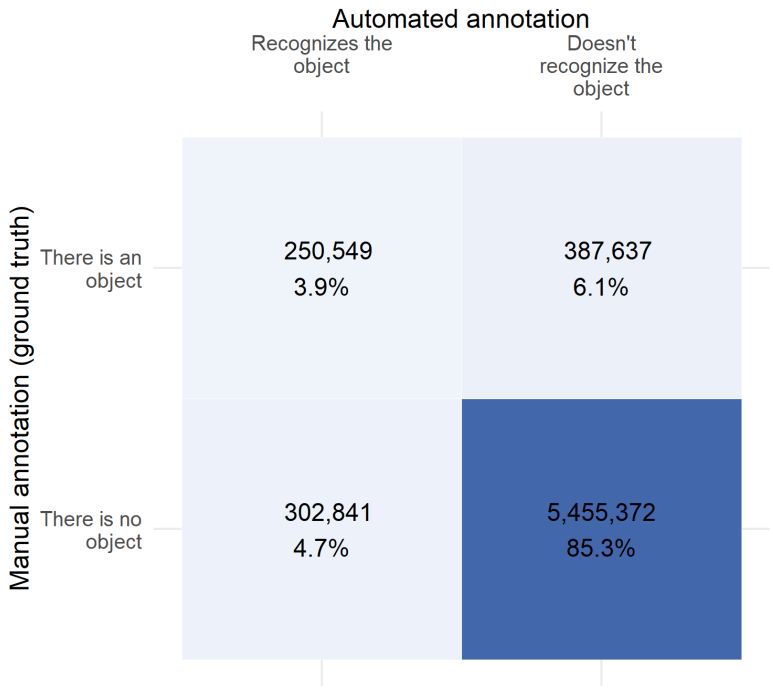



**Figure 1:** Confusion matrix of the caregiver showing the true positives, false negatives, false positives, and true negatives.

Child:

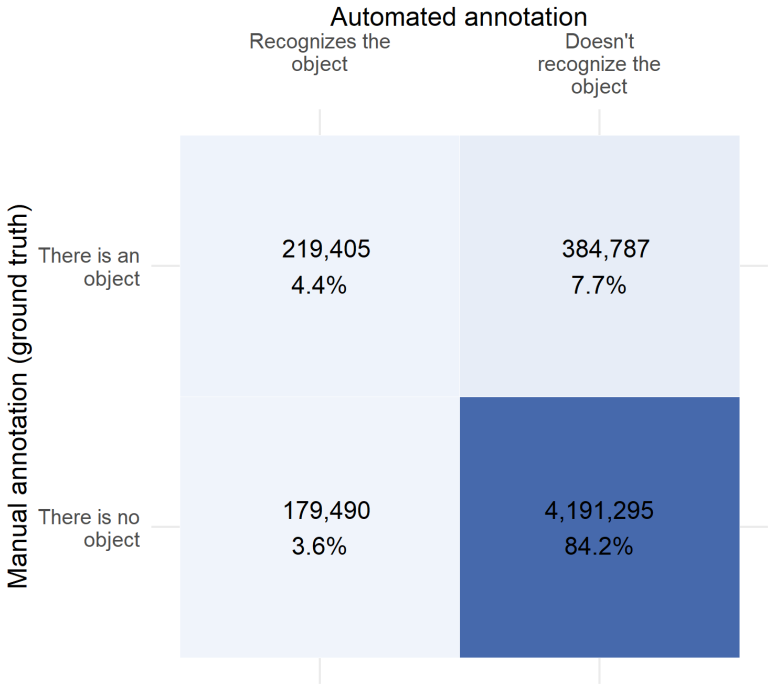



**Figure 2:** Confusion matrix of the child.

### *3.1.2 Dyad 2: P101S1*

Caregiver:

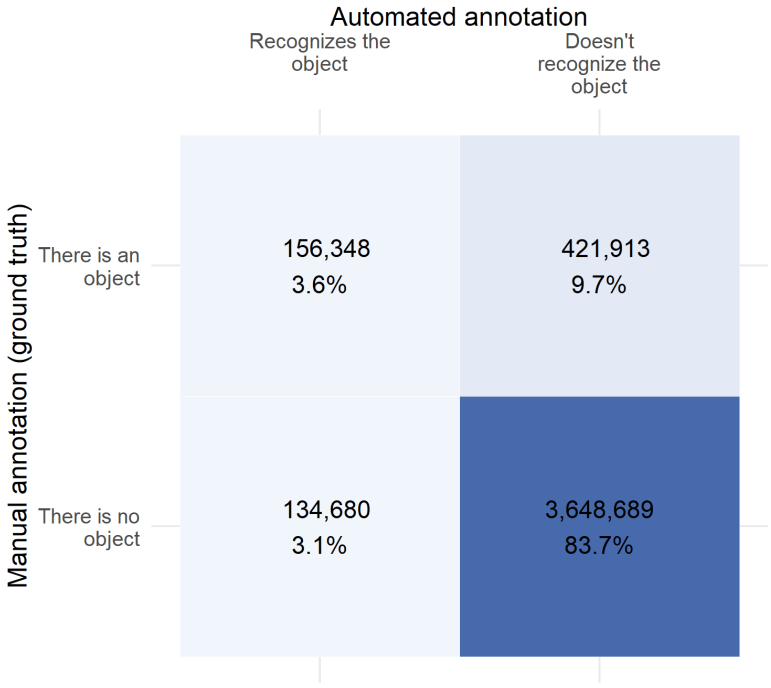



**Figure 3:** Confusion matrix of the caregiver showing the true positives, false negatives, false positives, and true negatives.

Child:

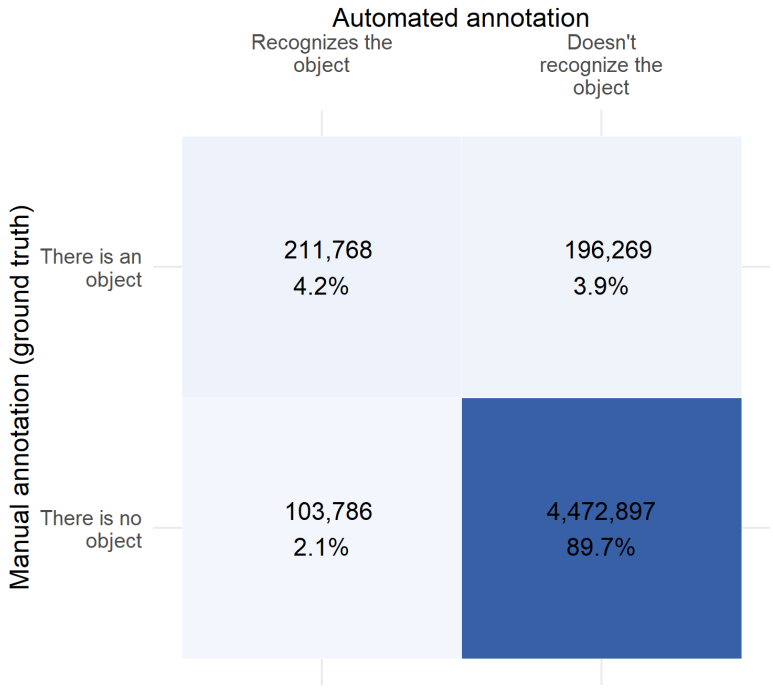



**Figure 4:** Confusion matrix of the child showing the true positives, false negatives, false positives, and true negatives.

#### *3.1.3 Metrics*

| | P06 caregiver | P06 child | P101 caregiver | P101 child |
|---|---|---|---|---|
| precision | 45.3 | 55.0 | 53.7 | 67.1 |
| recall | 39.3 | 36.3 | 27.0 | 51.9 |
| specificity | 94.7 | 95.9 | 96.4 | 97.7 |

**Table 2:** Precision, recall, and specificity % of automated annotation per participant

### 3.2 Object-level comparisons

#### *3.2.1 Dyad 1: P06S2*

Caregiver:

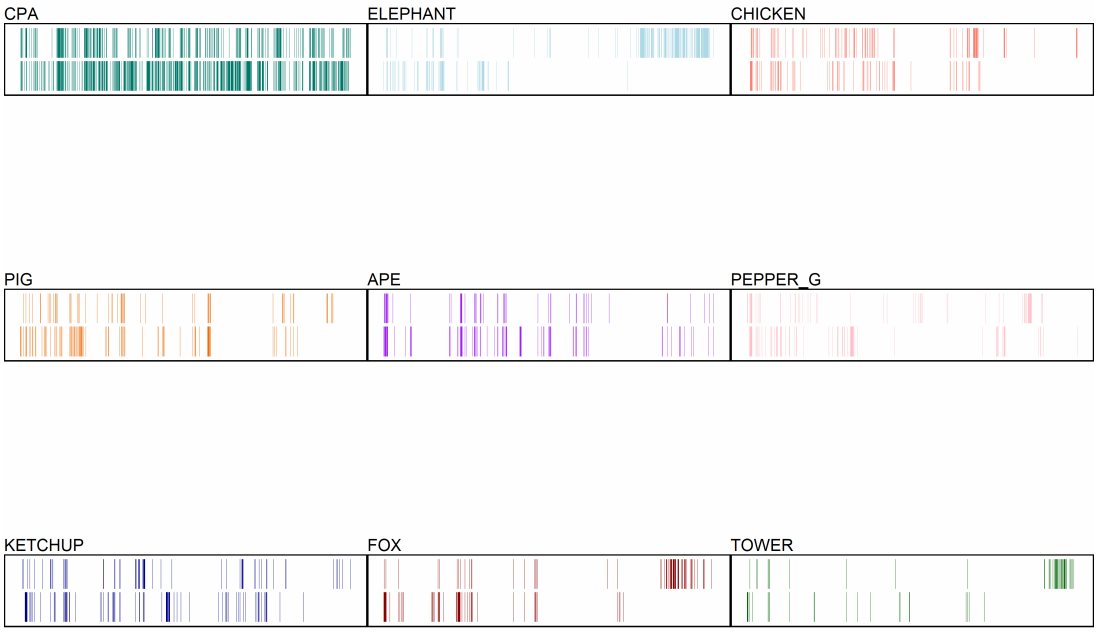



**Figure 5:** Object-level comparison of the caregiver. Upper stripes are the automated annotations, and bottom stripes are the ground-truth manual annotations.

Child:

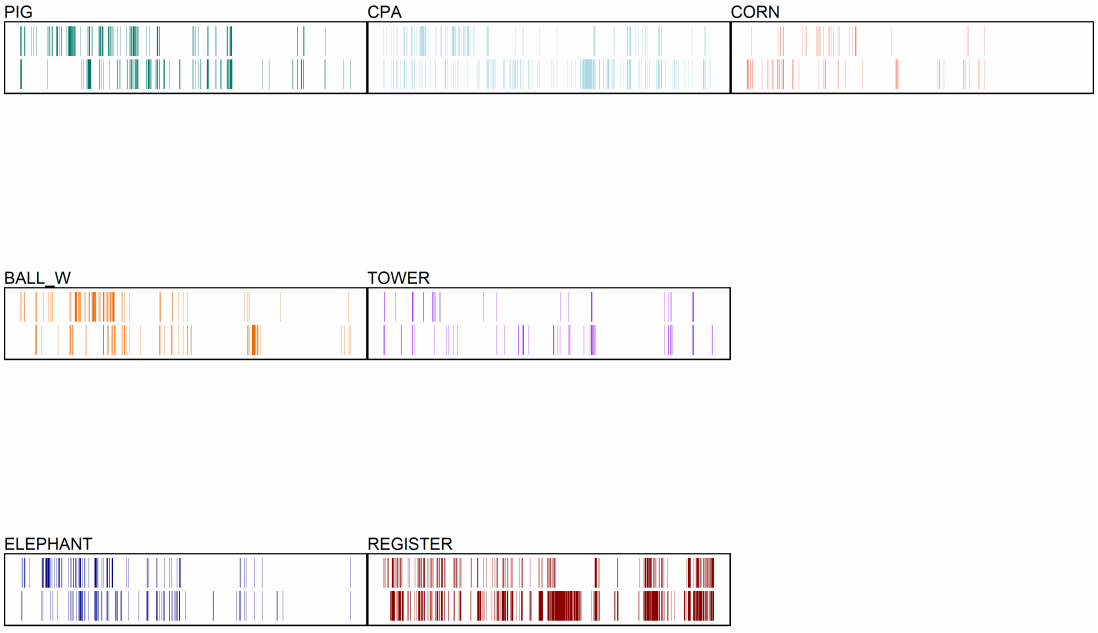



**Figure 6:** Object-level comparison of the child. Upper stripes are the automated annotations, and bottom stripes are the ground-truth manual annotations.

### *3.2.2 Dyad 2: P101S1*

Caregiver:

| Object | TP (%) | FP (%) | TN (%) | FN (%) | Precision (%) | Recall (%) |
|---|---|---|---|---|---|---|
| **Adult** | | | | | | |
| Co-Participant | 21.47 | 3.97 | 53.47 | 21.10 | 84.41 | 50.43 |
| Pig | 1.75 | 2.24 | 90.55 | 5.45 | 43.88 | 24.31 |
| Ketchup | 2.10 | 3.52 | 88.98 | 5.39 | 37.37 | 28.04 |
| Elephant | 1.55 | 14.06 | 79.47 | 4.92 | 9.96 | 24.03 |
| Ape | 2.93 | 0.72 | 93.25 | 3.10 | 80.26 | 48.62 |
| Fox | 1.56 | 4.77 | 90.33 | 3.34 | 24.65 | 31.83 |
| Chicken | 2.17 | 5.99 | 87.20 | 4.65 | 26.56 | 31.76 |
| Pepper green | 1.08 | 3.72 | 90.10 | 5.10 | 22.42 | 17.41 |
| Tower | 0.65 | 3.62 | 94.26 | 1.48 | 15.13 | 30.38 |
| **Child** | | | | | | |
| Pig | 3.96 | 5.65 | 85.60 | 4.80 | 41.19 | 45.21 |
| Ball white | 1.94 | 5.41 | 88.28 | 4.36 | 26.38 | 30.78 |
| Elephant | 3.43 | 3.83 | 87.89 | 4.85 | 47.28 | 41.44 |
| Co-Participant | 4.95 | 4.64 | 78.10 | 12.31 | 51.63 | 28.68 |
| Tower | 1.32 | 1.05 | 95.27 | 2.37 | 55.61 | 35.73 |
| Register | 14.49 | 2.46 | 61.55 | 21.50 | 85.49 | 40.27 |
| Corn | 0.78 | 2.21 | 93.05 | 3.95 | 26.08 | 16.50 |
| **Mean** | **4.13** | **4.24** | **84.83** | **6.79** | **42.39** | **32.84** |

**Table 3:** Per-object automated vs manual gaze comparison for the P06S2 dyad.

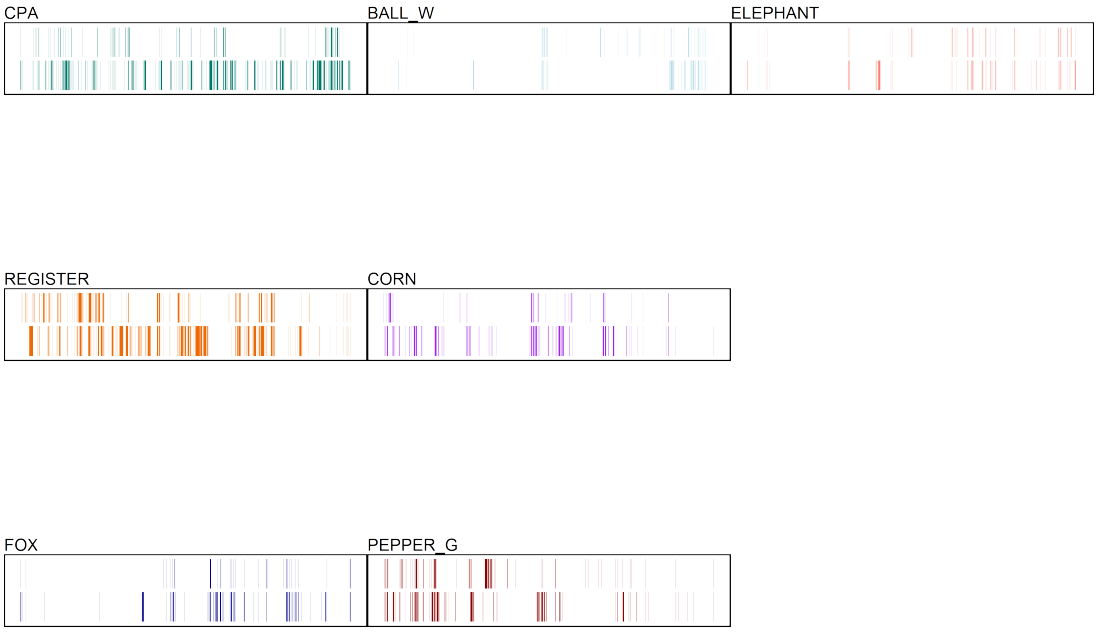



**Figure 7:** Object-level comparison of the caregiver. Upper stripes are the automated annotations, and bottom stripes are the ground-truth manual annotations.

Child:

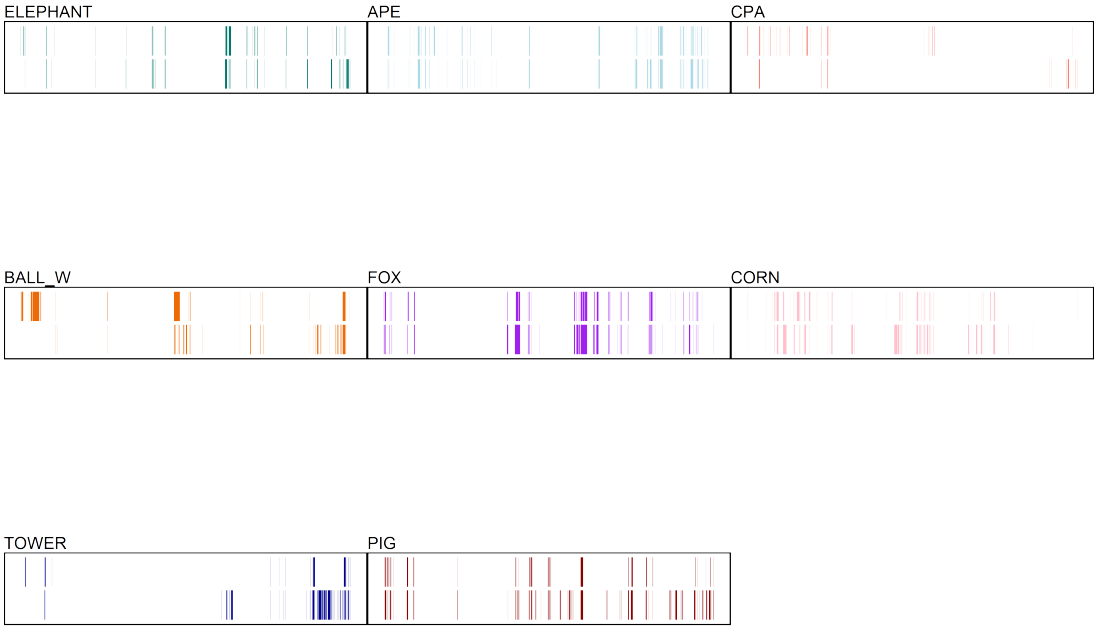



**Figure 8:** Object-level comparison of the child. Upper stripes are the automated annotations, and bottom stripes are the ground-truth manual annotations.

| Object | TP (%) | FP (%) | TN (%) | FN (%) | Precision (%) | Recall (%) |
|---|---|---|---|---|---|---|
| **Adult** | | | | | | |
| Co-Participant | 4.03 | 4.01 | 71.88 | 20.07 | 50.14 | 16.74 |
| Register | 9.43 | 5.79 | 66.95 | 17.83 | 61.97 | 34.58 |
| Fox | 2.62 | 0.91 | 90.91 | 5.55 | 74.24 | 32.11 |
| Ball white | 1.62 | 2.23 | 92.76 | 3.38 | 42.10 | 32.44 |
| Corn | 1.61 | 2.27 | 85.82 | 10.29 | 41.52 | 13.57 |
| Pepper green | 2.96 | 4.36 | 85.57 | 7.11 | 40.45 | 29.39 |
| Elephant | 2.81 | 2.04 | 91.67 | 3.48 | 57.88 | 44.66 |
| **Child** | | | | | | |
| Elephant | 3.17 | 1.42 | 92.92 | 2.48 | 69.04 | 56.10 |
| Ball white | 2.13 | 4.29 | 90.38 | 3.19 | 33.18 | 40.04 |
| Tower | 1.69 | 0.86 | 91.33 | 6.12 | 66.43 | 21.68 |
| Ape | 7.08 | 2.33 | 88.25 | 2.33 | 75.21 | 75.21 |
| Fox | 8.64 | 1.37 | 86.63 | 3.35 | 86.32 | 72.04 |
| Pig | 4.76 | 1.02 | 87.49 | 6.72 | 82.32 | 41.46 |
| Co-Participant | 0.73 | 2.94 | 95.10 | 1.23 | 19.87 | 37.11 |
| Corn | 5.77 | 2.42 | 85.74 | 6.06 | 70.45 | 48.78 |
| **Mean** | **3.94** | **2.55** | **86.90** | **6.61** | **58.07** | **39.73** |

**Table 4:** Per-object automated-vs-manual gaze annotation comparison for the P101S1 dyad.